

1. Motivation

I am generally interested in understanding the reasoning and evidence behind rules or assumptions that are widely accepted over time. Rather than taking such ideas for granted, I often question whether they remain valid under different conditions and datasets.

One of the questions that motivated this project was whether alcohol consumption is truly associated with higher traffic fatality rates at the macro level. While the dangers of driving under the influence are well established at the individual level, I became interested in examining whether this relationship remains clearly observable when comparing countries with different social and economic conditions.

This naturally extended to broader socio-economic questions, such as whether higher GDP per capita is consistently associated with safer roads. By utilizing data-driven methodologies, this project aims to analyze and predict traffic fatality rates using non-infrastructure, macro-level socio-economic indicators.

2. Data Source & Feature Engineering

2.1 Data Acquisition and Variable Descriptions

To conduct this macro-level analysis, data was aggregated from two highly reputable global databases to construct a unified 20-year panel dataset spanning from 2000 to 2019. The datasets were aligned and merged on the country-year level using Python's pandas library.

- ***GDP_Per_Capita***: Source: [World Bank \(NY.GDP.PCAP.CD\)](#). Gross Domestic Product per capita in current US dollars. Acts as a primary proxy for national economic development and infrastructural capacity.
- ***Alcohol_Per_Capita***: Source: [World Bank \(SH.ALC.PCAP.LI\)](#). Total annual consumption (in liters of pure alcohol) per capita for ages 15 and older. Captures broad cultural and behavioral consumption patterns.
- ***Population***: Source: [World Bank \(SP.POP.TOTL\)](#). Total national population. Used primarily to evaluate whether population size introduced scale-related bias into the predictive models.

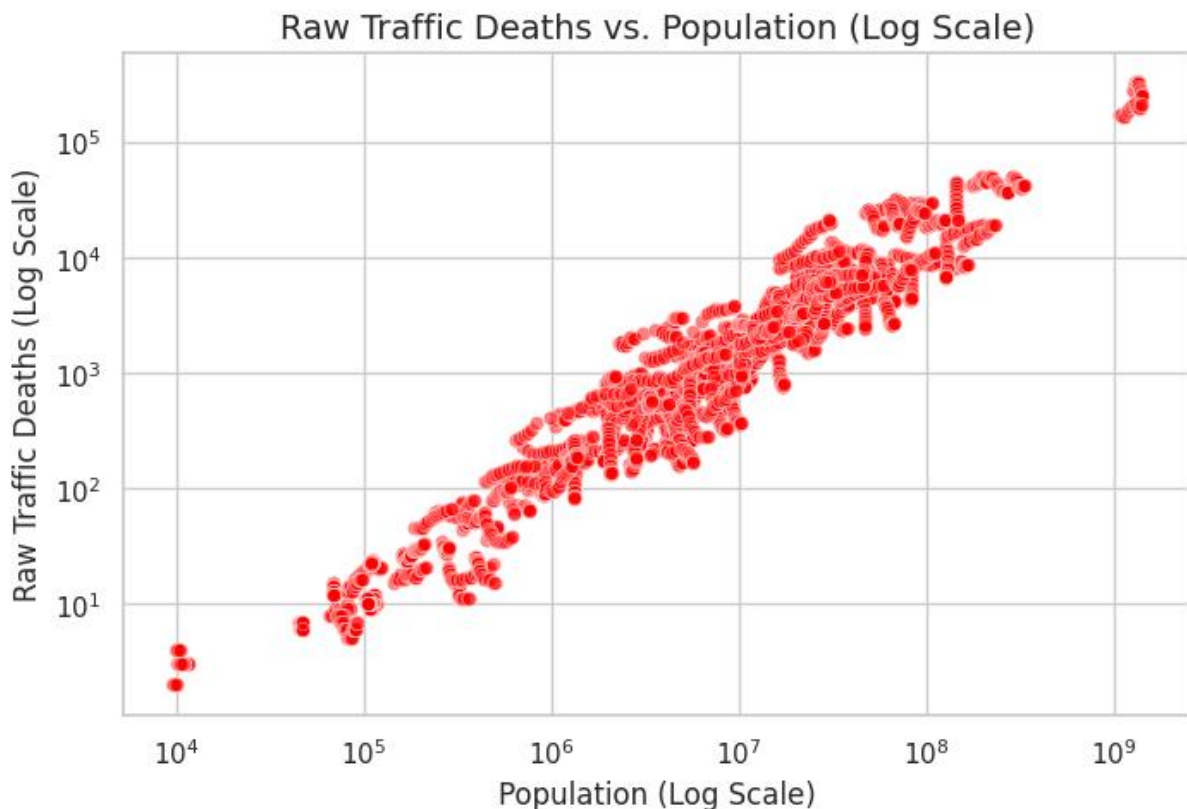
- **Total_Traffic_Fatalities:** Source: [Kaggle Dataset](#). The raw, absolute number of estimated road traffic deaths per country annually.

Note: The raw fatality counts are not used directly in the analysis. Instead, normalized values are used to allow more meaningful comparisons between countries, as explained in the next section.

2.2 Rationale for Population Normalization (Per 100k)

Instead of using the raw *Total_Traffic_Fatalities* values directly, a normalized target variable, **Death_Rate_Per_100k**, was derived by dividing the total number of fatalities by the national population and multiplying the result by 100,000.

Population distributions across countries exhibit substantial variance, ranging from small island states to nations with populations exceeding one billion. Incorporating raw population and fatality counts directly into the predictive model could introduce significant scale-related bias. Under such conditions, the model may disproportionately rely on population magnitude to estimate traffic fatalities, rather than learning the underlying socio-economic, infrastructural, and behavioral factors associated with road safety risk.



As illustrated in the exploratory visualizations, transforming absolute figures into a standardized metric (per 100,000 people) substantially reduces the population scale bias. This normalization isolates the true impact of wealth (*GDP_Per_Capita*) and behavior (*Alcohol_Per_Capita*), ensuring that the resulting predictive models evaluate safety outcomes relative to population density and exposure, rather than absolute national magnitude.

3. Data Analysis Pipeline

The analytical architecture of the project was executed in three progressive stages:

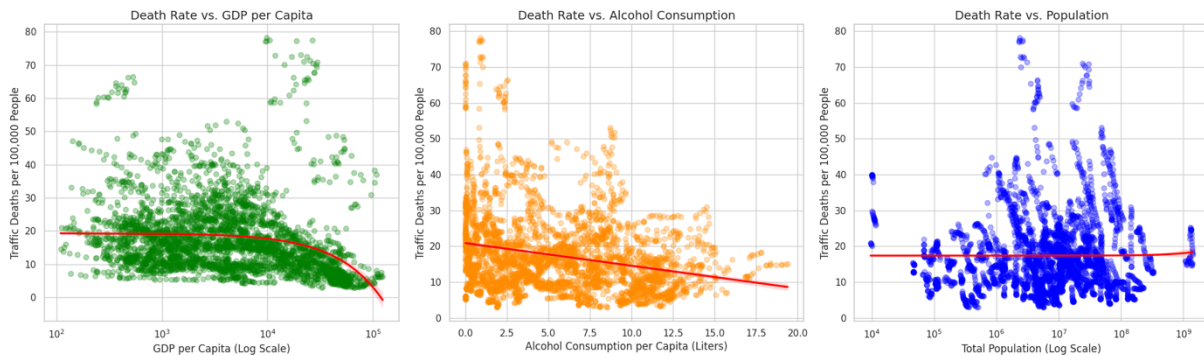
1. **Exploratory Data Analysis (EDA):** Initial statistical analysis revealed extreme right-skewness in global wealth distributions, indicating that a minority of high-income nations form a long tail. Correlation matrices and scatter plots were utilized to identify preliminary feature associations and uneven variance patterns in the data.
2. **Hypothesis Testing:** Statistical tests were applied to validate the statistical significance ($P\text{-value} < 0.05$) of observed socio-economic correlations, ensuring that differences in road safety between distinct income brackets were not random occurrences.
3. **Machine Learning:** Because the dataset exhibited severe non-linear interactions, parametric models like Multiple Linear Regression performed poorly. Consequently, the project transitioned to a non-linear, tree-based ensemble method: the Random Forest Regressor. The model was optimized via GridSearchCV for hyperparameter tuning and validated using 5-Fold Cross-Validation to minimize the risk of overfitting. Finally, an additional Sensitivity Analysis was applied by excluding the Population feature to ensure the model learned socio-economic drivers rather than memorizing country sizes.

4. Key Empirical Findings

4.1 The Behavioral Paradox and Correlation Analysis

Highly developed nations, particularly in Western Europe, tend to report higher levels of alcohol consumption while also generally exhibiting stronger road infrastructure, stricter traffic regulations, and more advanced emergency

healthcare systems. Although these factors were not directly measured in the dataset, they may partially explain why higher alcohol consumption does not necessarily correspond to higher traffic fatality rates at the macro level.

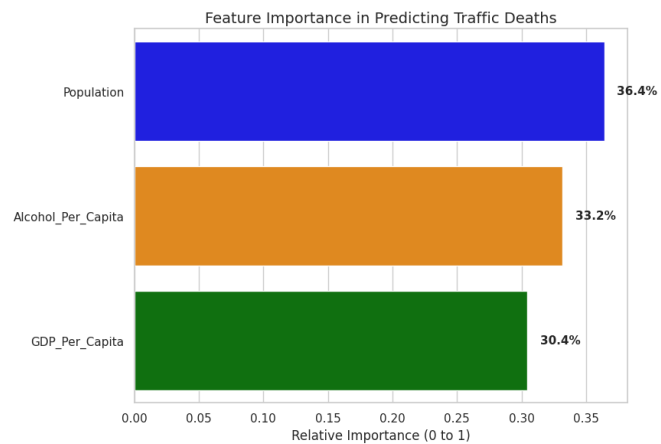
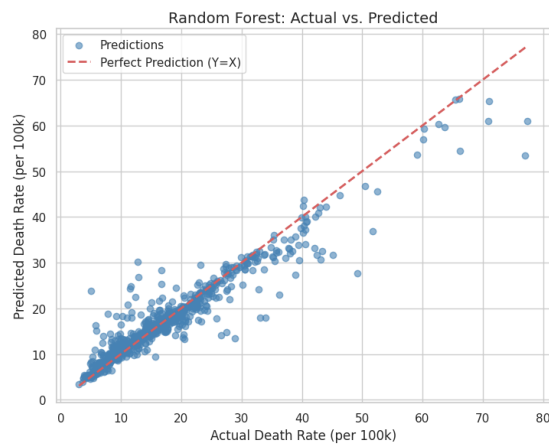


As the correlation visualizations demonstrate, the relationship between alcohol consumption and traffic fatality rates at the macro level is likely confounded by broader structural factors. Higher alcohol consumption may be associated with countries that also have more developed infrastructure, stricter traffic enforcement, and better healthcare systems, although this relationship is not directly measured in the dataset. As a result, alcohol consumption in this dataset may not directly reflect reckless driving behavior but instead may act as a proxy for a country's socio-economic development and regional characteristics, although this remains speculative.

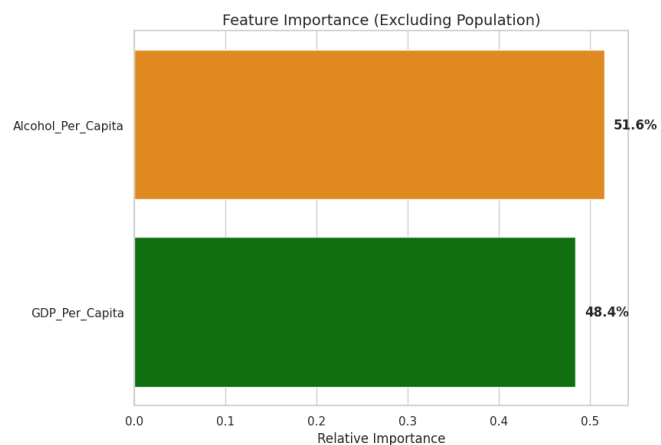
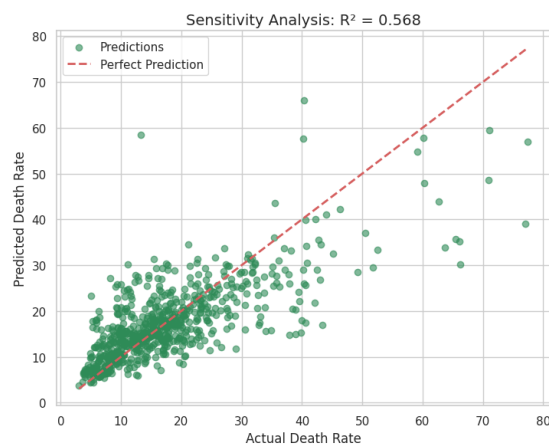
It should also be noted that the observed correlation between GDP per capita and alcohol consumption is only moderate (approximately 0.38). This suggests that alcohol consumption patterns are shaped not only by economic conditions but also by cultural, religious, and regional factors that are not explicitly captured in the dataset.

4.2 Predictive Power and Feature Importance

In all machine learning modeling iterations, the Random Forest algorithm demonstrated strong performance in capturing the non-linear relationships within the data. As visualized in the model evaluation charts, the algorithm successfully maps the complex socio-economic inputs to the actual traffic fatality rates with minimal variance ($R^2 \approx 0.9$).



The feature importance analysis further corroborates the exploratory findings. The algorithm consistently selected *Alcohol_Per_Capita* as the most important feature. This finding suggests that behavioral patterns acting as sensitive proxies for regional patterns of socio-economic development have strong predictive power in estimating fatality risks, even more so than raw economic output alone.



4.3 Model Performance Matrix:

- Primary Random Forest Model (Using GDP, Alcohol, and Population): $R^2 = 0.9031$
- Sensitivity Analysis (Robustness check excluding Population): $R^2 = 0.5679$

(Note: Explaining over 56% of the global variance in traffic fatalities using only two features Wealth and Alcohol supports the central hypothesis without relying on population-based mathematical shortcuts.)

5. Limitations and Future Work

While the models demonstrated strong predictive performance, several limitations should be noted. First, the study relies entirely on national-level aggregated data, which introduces the risk of ecological fallacy, where country-level averages may hide important regional or demographic differences within countries. As a result, patterns observed at the national level may not accurately reflect local variations in road safety outcomes.

Second, the feature set does not include detailed infrastructural or policy-level variables. Factors such as the strictness of DUI (Driving Under the Influence) laws, the density of traffic enforcement cameras, road quality, and other local safety measures are not explicitly captured in the dataset, which limits the model's ability to fully represent real-world driving conditions.

Future work could address these limitations by incorporating more granular city-level or regional data, as well as direct public policy indicators. This would enable the development of models better suited to identifying causal relationships and would allow for a more precise evaluation of the impact of specific traffic safety regulations over time.

6. Usage of AI and LLMs

In accordance with the DSA210 course guidelines, AI tools and large language models (LLMs) were used strictly in a supportive role. Their use was limited to the following areas:

- **Code Optimization:** AI was used to assist with code debugging and to provide suggestions for resolving implementation and data-processing issues encountered during the project.
- **Documentation Refinement:** AI was used for language refinement, grammatical corrections and improving the academic presentation of the markdown notebooks.

All core research questions, modelling choices, and analytical interpretations were conducted manually and remain the original work of the author.